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Problem Set 2

Mathematics for Social Scientists

Exercise 1. *(Essential Mathematics for Economic Analysis (5th edition),* Knut Sydsaeter and Peter Hammond, Section 1.2, Question 2)
Determine which of the following formulas are true. If any formula is false, find a counter example to demonstrate this, using a Venn diagram if you find it helpful.

a. $A \subseteq B \iff A \cup B = B$

b. $A \subseteq B \iff A \cap B = A$

c. $A \cap B = A \cap C \Rightarrow B = C$

d. $A \cup B = A \cup C \Rightarrow B = C$

e. $A = B \iff (x \in A \iff x \in B)$

Exercise 2. *(Essential mathematics for Economic Analysis (5th edition),* Knut Sydsaeter and Peter Hammond, Section 5.3, Question 7)
If f is the function that tells you how many kilograms of carrots you can buy for a specified amount of money, then what does f^{-1} tell you?

Exercise 3. (Wherever they exist) find the inverse of the following functions:

a. $y = 1/x^2$

b. $y = \sqrt{\sqrt{x} - 5}$

c. $y = 3^x$

d. $y = x^2 + 2x + 1$

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Exercise 4. Are the following functions injective/surjective/bijective?

(i) $f_1 : \{1,2,3\} \to \{1,2,3,4,5\}$ with $f_1(1) = 2$, $f_1(2) = 1$ and $f_1(3) = 5$

(ii) $f_2 : \mathbb{R} \to \mathbb{R} : x \to x^2$

(iii) $f_3 : \mathbb{R} \to \mathbb{R}_0^+ : x \to x^2$

(iv) $f_4 : \mathbb{Z} \to \mathbb{Z} : n \to n + 1$

(v) $f_5 : \mathbb{R} \to \mathbb{R} : x \to x^3$

(vi) $f_6 : \mathbb{R}_0^+ \to \mathbb{R} : x \to x^3$

(vii) $f_7 : (0, \infty) \to (0, \infty) : x \to \frac{1}{x}$

Notation: $\mathbb{R}_0^+ \coloneqq \{x \in \mathbb{R} : x \geq 0\}$.

Exercise 5. Find the following limits (**without** using the L'Hospital's Rule):

Here is an example:

$$\lim_{x\to\infty} \frac{2x^3+3}{x^3-5} = \lim_{x\to\infty} \frac{2+\frac{3}{x^3}}{1-\frac{5}{x^3}} = \frac{\lim_{x\to\infty} 2+\frac{3}{x^3}}{\lim_{x\to\infty} 1-\frac{5}{x^3}} = \frac{2}{1} = 2$$

(i) $\lim_{x\to4}[x^2 - 3x + 4]$

(ii) $\lim_{x\to4} \frac{x^2}{3x-5}$

(iii) Given $\lim_{y\to0} f(y) = -2$ and $lim_{y\to0} g(y) = 2$, find $\lim_{y\to0}[f(y) - 3f(y)g(y)]$.

(iv) $\lim_{x\to\infty} \frac{11x^2}{x^2+3}$

(v) $\lim_{x\to\infty} \frac{3x-12}{x+1}$

(vi) $\lim_{x\to\infty} \frac{2^x-3}{2^x+5}$

Exercise 6. Find the following limits **without** using L'Hôpital's Rule.

(i) $\displaystyle\lim_{t\to-2} \frac{t^3+8}{t+2}$

(ii) $\displaystyle\lim_{t\to-1} \frac{t^3+t^2-5t+3}{t^2-3t+2}$

(iii) $\displaystyle\lim_{x\to4} \frac{x^2-16}{x-4}$

(iv) $\displaystyle\lim_{x\to2} \frac{x^2-4x+4}{x^2+x-6}$

(v) $\displaystyle\lim_{t\to1} \frac{t^3+t^2-5t+3}{t^3-3t+2}$

(vi) $\displaystyle\lim_{x\to3^+} \frac{x}{x-3}$

(vii) $\displaystyle\lim_{x\to1^+} \frac{x^4-1}{x-1}$

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(viii) $\lim_{x\to 4^-} \dfrac{3-x}{x^2-2x-8}$ and $\lim_{x\to 4^+} \dfrac{3-x}{x^2-2x-8}$ (x) $\lim_{x\to\infty} \dfrac{3x+1}{2x-5}$

(ix) $\lim_{y\to 4} \dfrac{4-y}{2-\sqrt{y}}$

Exercise 7. Find the limits

- $\lim_{n\to\infty} a_n$,
- $\lim_{n\to\infty} b_n$,
- $\lim_{n\to\infty} a_n/b_n$, (make sure this is well-defined)

for the following sequences $(a_n)_{n\in n}, (b_n)_{n\in n}$.

(i) $a_n = n$, $b_n = n^2$

(ii) $a_n = 2^n, b_n = 3^n$

(iii) $a_n = \frac{2n+2}{n+1}$, $b_n = \frac{n^2-1}{n^2+2n}$

(iv) $a_n = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right)$, $b_n = 1 + \frac{1}{n}$
Hint: the sequence $(a_n)_{n\in n}$ is called a *telescoping sum.*

Exercise 8. Let f and g be functions $\mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 - 1, & \text{for } x \leq 0, \\ -x^2, & \text{for } x > 0. \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} 3x - 2, & \text{for } x \leq 2, \\ -x + 6, & \text{for } x > 2. \end{cases}$$

Draw a graph of each function. Is f continuous at $x = 0$? Is g continuous at $x = 2$?

Exercise 9. For what value of a is the following function $f : \mathbb{R} \to \mathbb{R}$ continuous for all x?

$$f(x) = \begin{cases} ax - 1, & \text{for } x \leq 1, \\ 3x^2 + 1, & \text{for } x > 1. \end{cases}$$